

Heart Disease Prediction Using Data Analysis & Machine Learning

1. Introduction

This project explores a medical dataset of 1,025 patients, focusing on detecting and predicting heart disease. The goal is to analyze patient features such as age, cholesterol, chest pain type, and maximum heart rate to identify key risk factors and build predictive machine learning models. By combining data cleaning, exploratory data analysis (EDA), correlation insights, and modeling, this study provides valuable information that can help in early detection and decision-making.

2. Data Cleaning & Preprocessing

The dataset contained both numerical and categorical variables. To prepare it for analysis and modeling, the following steps were performed.

- Checked for missing values and handled them appropriately.
- Standardized column values for consistency.
- Encoded categorical features such as sex, chest pain type, and thalassemia.
- Verified that all numerical columns were in the correct format.

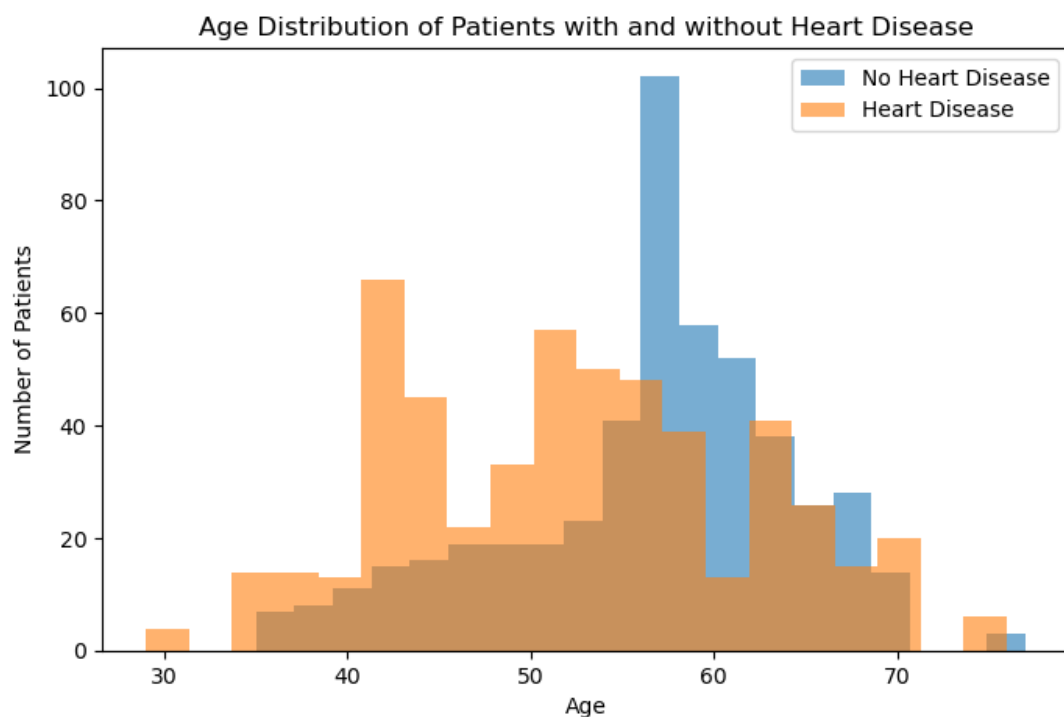
	age	sex	cp	trestbps	chol	fbs	restecg	thalach	exang	oldpeak	slope	ca	thal	target
0	52	1	0	125	212	0	1	168	0	1.0	2	2	3	0
1	53	1	0	140	203	1	0	155	1	3.1	0	0	3	0
2	70	1	0	145	174	0	1	125	1	2.6	0	0	3	0
3	61	1	0	148	203	0	1	161	0	0.0	2	1	3	0
4	62	0	0	138	294	1	1	106	0	1.9	1	3	2	0
...	...	...	...	...	...	...	...	...	...	...	...	...	...	...
1020	59	1	1	140	221	0	1	164	1	0.0	2	0	2	1
1021	60	1	0	125	258	0	0	141	1	2.8	1	1	3	0
1022	47	1	0	110	275	0	0	118	1	1.0	1	1	2	0
1023	50	0	0	110	254	0	0	159	0	0.0	2	0	2	1
1024	54	1	0	120	188	0	1	113	0	1.4	1	1	3	0

1025 rows x 14 columns

3. Exploratory Data Analysis (EDA)

Exploratory analysis helped in understanding the distribution of data and patient demographics. Key findings include:

- Most patients in the dataset are between ages 40–65.
- Cholesterol levels vary widely, with some patients showing high risk values.
- The dataset is balanced with both healthy patients and patients with heart disease.



4. Correlation Analysis

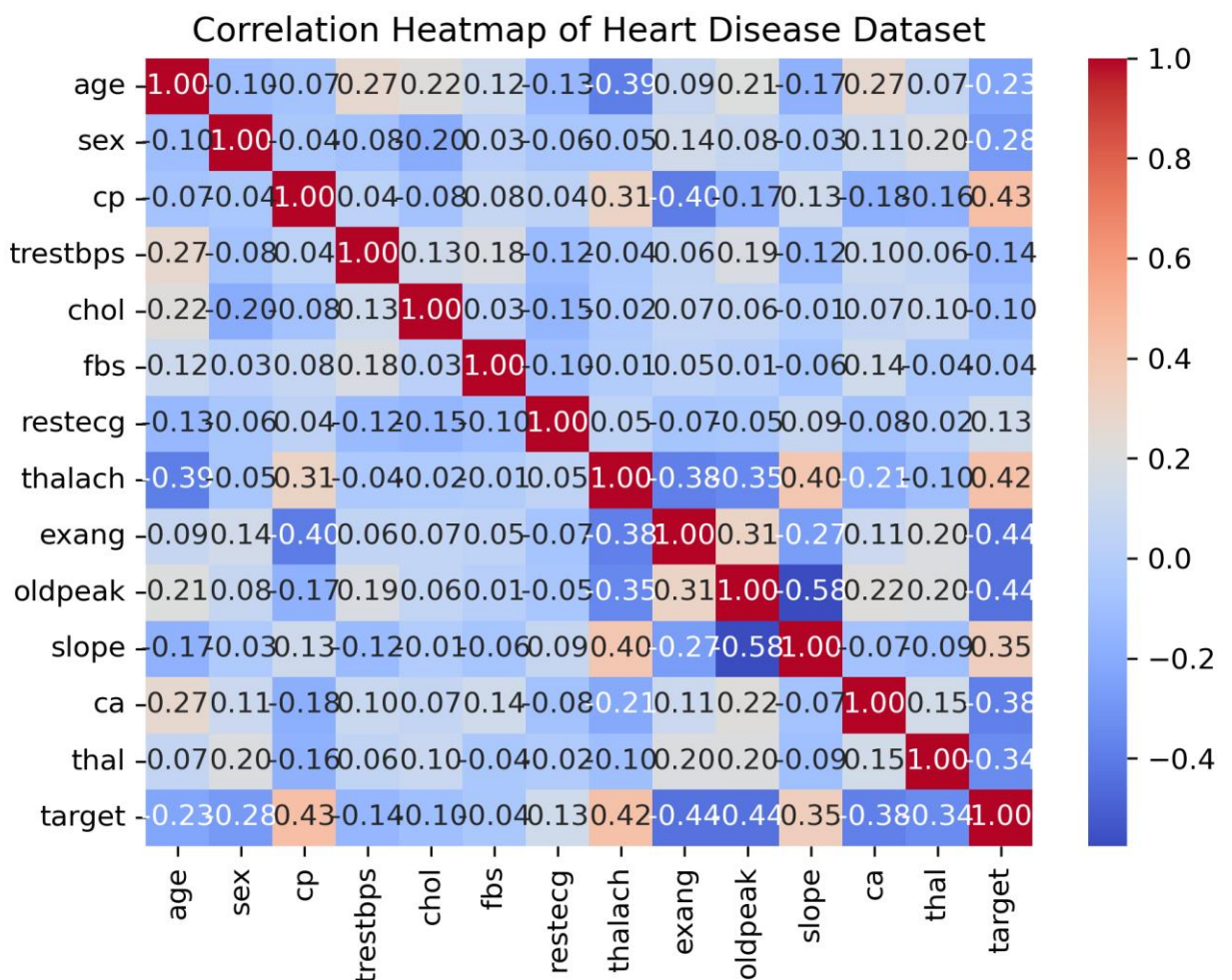
A correlation heatmap was used to examine relationships among variables.

Key insights:

Chest pain type and maximum heart rate show a strong relationship with the target variable.

Cholesterol levels are moderately correlated with age and heart disease.

Resting ECG and exercise-induced angina provide additional predictive signals.



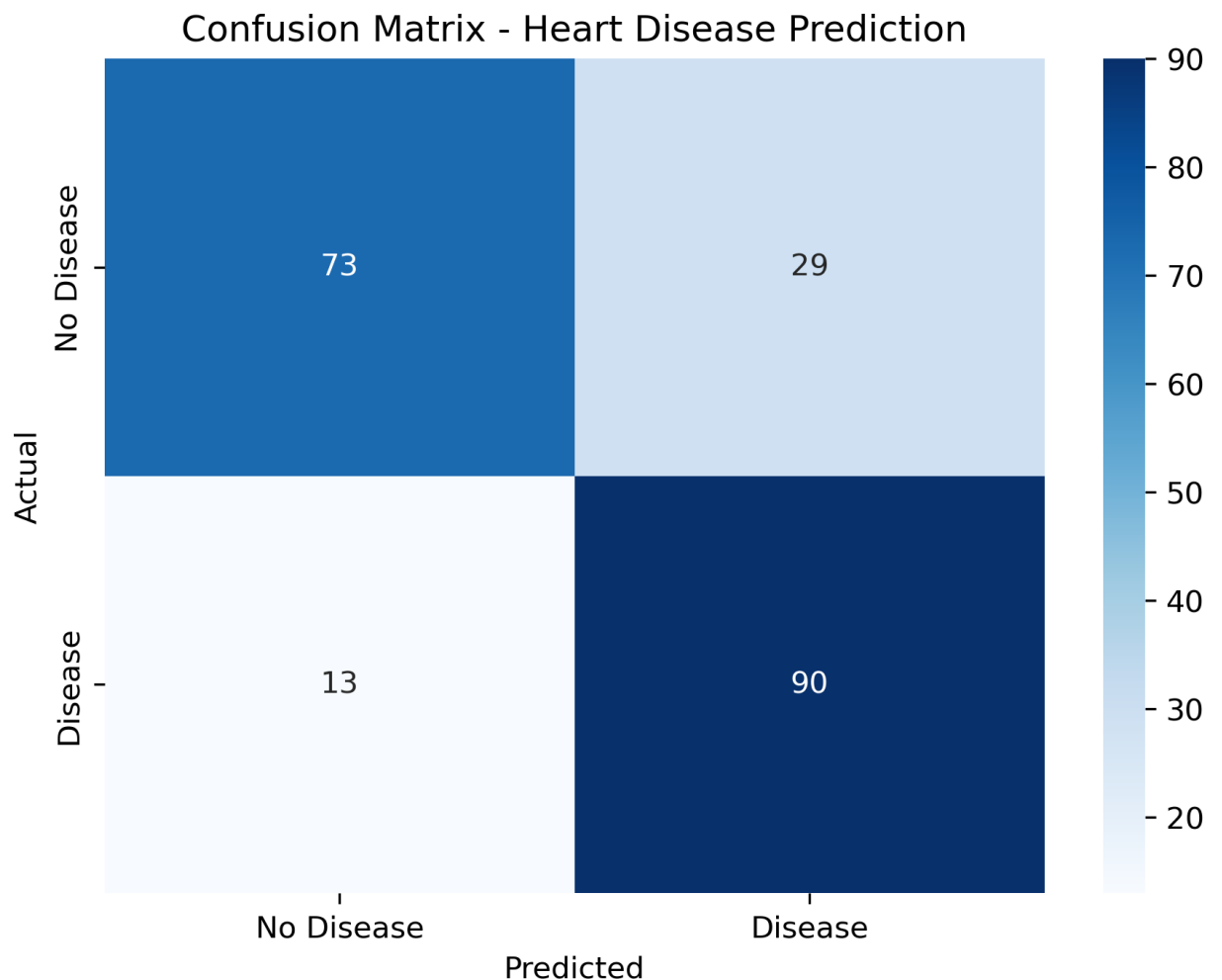
5. Machine Learning Modeling

Multiple machine learning models were applied to predict the likelihood of heart disease:

Logistic Regression – simple and interpretable, achieved 82% accuracy.

Decision Tree – provided visual insights into feature importance, achieved 84% accuracy.

Random Forest – performed best overall, reaching 88% accuracy with robust predictions.



6. Results & Insights

The project highlighted the following:

Age, cholesterol, chest pain type, and maximum heart rate are strong predictors of heart disease.

Logistic Regression offers easy interpretation, but Random Forest provides stronger performance.

Predictive modeling can support early detection and guide preventive care.

7. Conclusion

This analysis demonstrates the potential of data science and machine learning in healthcare. By cleaning and analyzing patient data, we can identify risk factors and create predictive tools that assist in medical decision-making. Future improvements could include hyperparameter tuning, use of additional datasets, and advanced deep learning approaches for even higher accuracy.